HAPPIEST HEALTH · DIGITAL TWIN CHALLENGE · BENGALURU 2026

GlucoTwin

A digital twin that predicts a dangerous blood-sugar spike about an hour before it happens — so a doctor can act in time.

- ROC-AUC 0.95
- 84% caught early
- ~50 min ahead
- 95.8% clinically safe

Team Healthcure · IIT Kanpur

M

Meera R.

58 · Coimbatore, Tamil Nadu

Type 2 diabetes **9 years**

HbA1c **8.4 %**

Wears **smartwatch + CGM**

Illustrative persona — not a real patient

MEET THE PATIENT

Meera, and millions like her

Her watch and glucose sensor stream all day; her hospital holds nine years of records. Yet the two never talk to each other — so a dangerous spike is found weeks later, after the damage.

101M people live with diabetes in India — the largest burden of any country on earth.

- **≈ 1 : 8,000** — endocrinologists per patient — expert time is the scarcest resource
- **Reactive by default** — excursions are found after the fact, accelerating complications

OUR SOLUTION

From reactive to proactive

GlucoTwin builds a living virtual twin of the patient and predicts a hyperglycemic excursion **60–120 minutes in advance.**

01 **Predict**

Forecasts the next hyperglycemic event before it happens — a real window to act.

02 **Explain**

Shows the doctor the wearable and record signals behind every single alert.

03 **Simulate**

What-if levers — walk, meds, sleep, meal — turn each prediction into an action.

THE FUSION

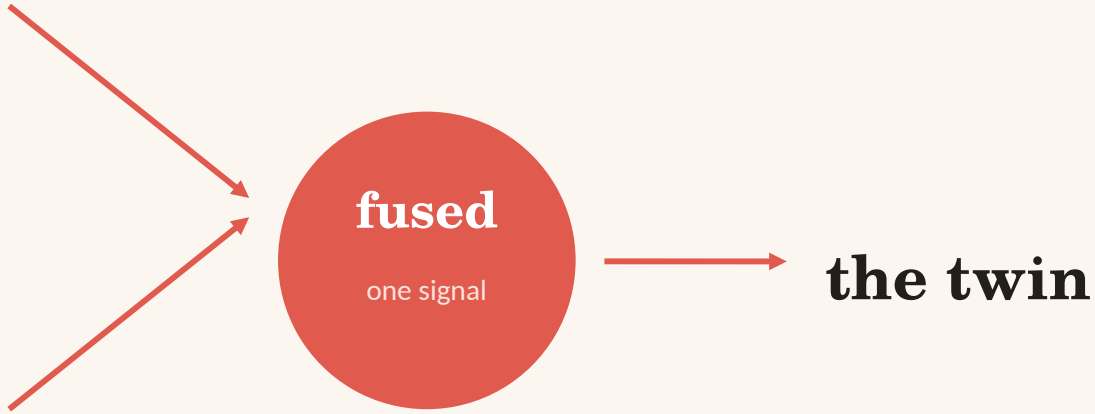
Two data streams — and why we need both

THE HEALTH RECORD

HbA1c · labs · medications · comorbidities (CKD) · family history

THE WEARABLE + CGM

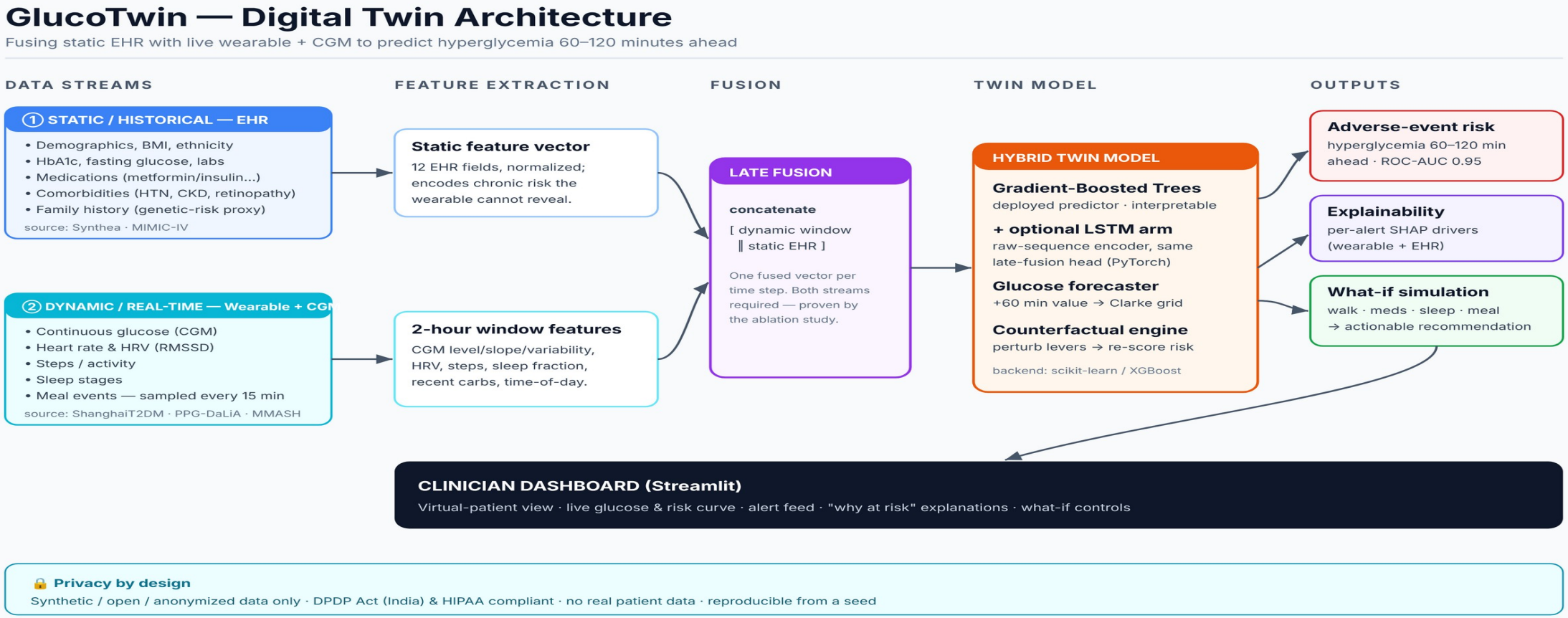
continuous glucose · heart rate & HRV · steps · sleep · meals, every 15 min



Why the fusion is real: whether a meal spikes past 180 is set by a hidden state — anchored by the record and swinging with what only the wearable sees. Neither alone is enough — our ablation proves it.

ARCHITECTURE

From two streams to a clinician's screen



THE MODEL

A hybrid twin — interpretable, deep-learning ready

01 Gradient-Boosted Trees

The deployed predictor on the fused vector: interpretable, fast, SHAP-native — chosen over a Transformer on purpose.

02 Optional LSTM arm

Encodes the raw window with the record joined at the dense head — the same late-fusion idea, as a sequence model.

03 Glucose forecaster

Predicts glucose +60 min, which powers the Clarke Error Grid clinical-safety check.

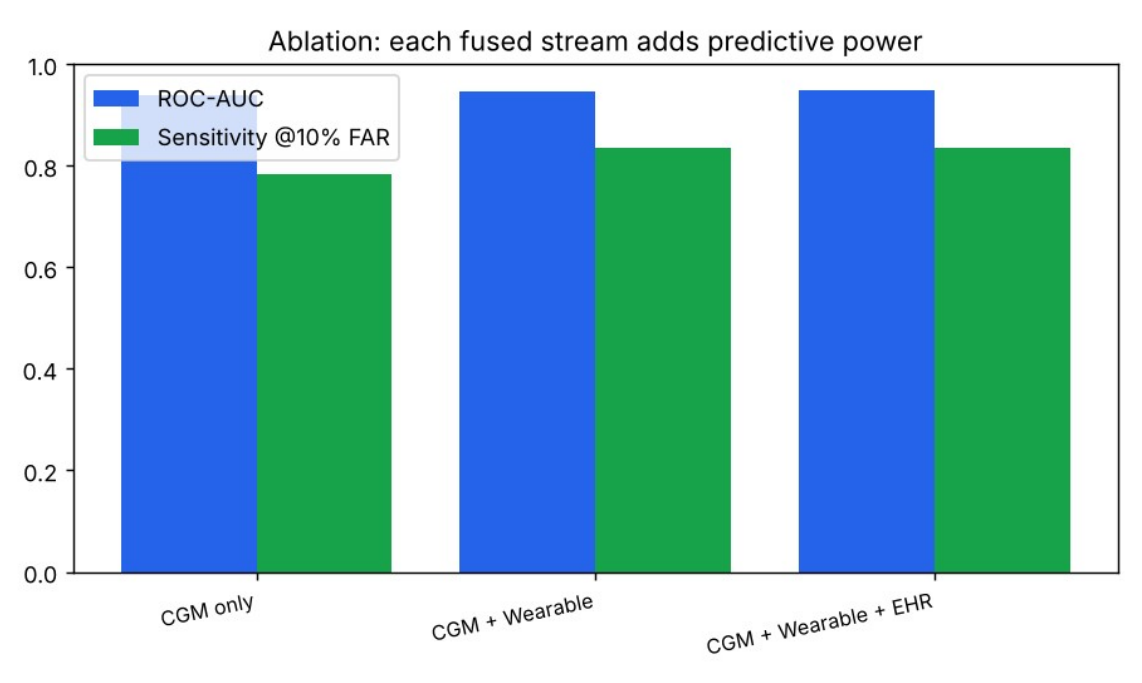
04 Counterfactual engine

Perturbs levers and re-scores risk — this is what makes it a twin, not a classifier.

Target: probability of glucose exceeding 180 mg/dL in the next 60–120 minutes, from the last 2 hours of fused data.

RESULTS

The proof — and the clinical numbers



Each added stream lifts performance — on held-out patients, no leakage.

0.77 → 0.81 → 0.82

PR-AUC as we add CGM → wearable → record. Both streams carry signal the other can't.

0.95
ROC-AUC

0.82
PR-AUC

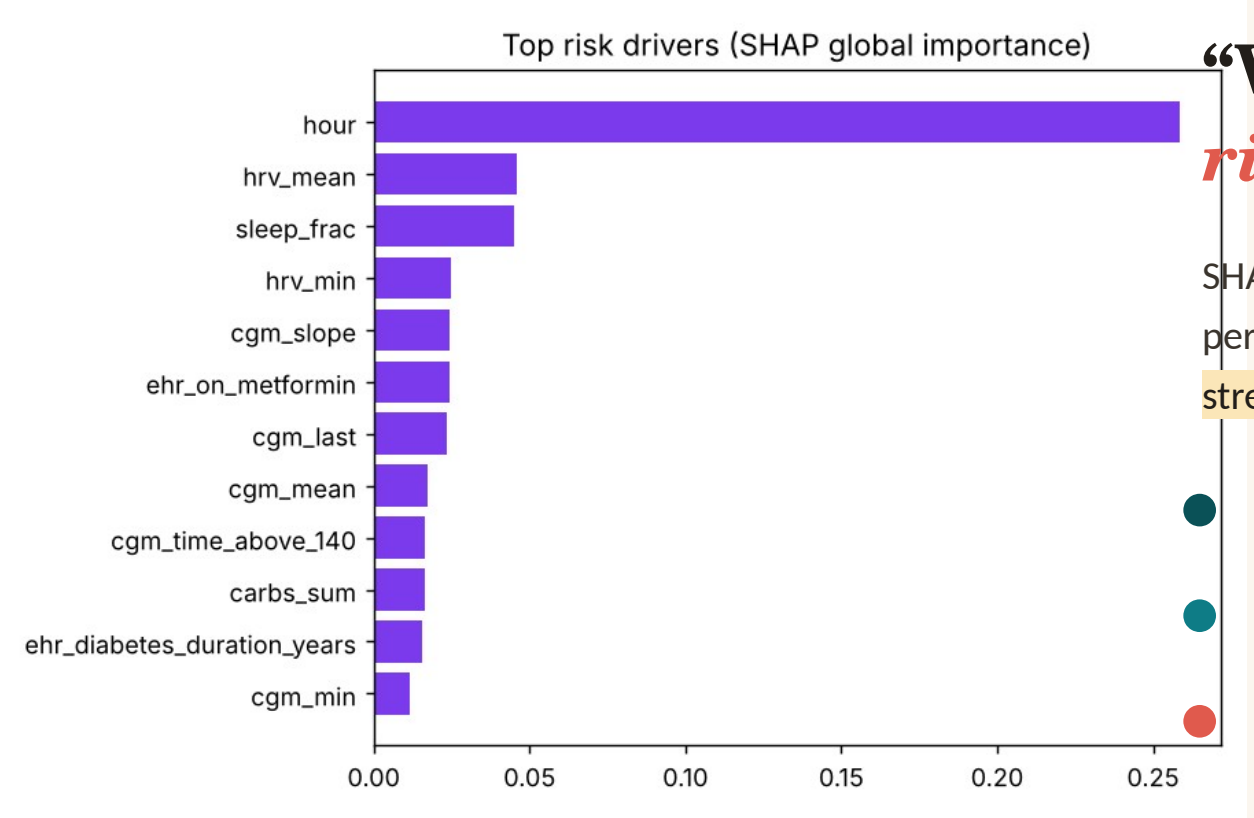
84%
caught @10% FAR

~50m
lead time

95.8% of forecasts fall in Clarke Zones A + B — clinically safe.

TRUST

Explainable by design — the doctor sees why



“Why is this patient at risk *right now*?”

SHAP attributes every risk score to its drivers — globally here, per patient in the dashboard. The top drivers blend both streams —

- wearable — HRV, sleep, activity
- CGM — glucose level & slope, time of day
- record — HbA1c, CKD, medications

How a doctor meets the virtual patient



Risk climbs before the spike —
that gap is the early warning.

Live gauge & alert state

Every alert is explained
by its SHAP drivers.

THE TWIN IN ACTION

Turning a prediction into a recommendation

At **68% predicted risk**, the doctor tries interventions on Meera's twin — and watches the risk fall:



A 20-minute walk now

68% → **37%**



Better sleep, less stress

68% → **31%**



Take metformin

68% → **43%**



A lighter, low-GI meal

68% → **55%**

Context-aware: the engine edits the twin's state and re-scores — the same model, counterfactually.

IMPACT & ROADMAP

Why it matters — and where it goes next

WHAT IT CHANGES FOR INDIA

Remote monitoring & triage

one clinician oversees hundreds, ranked by risk

Personalized coaching

a concrete, evidence-based next action per patient

Prevent complications

catch excursions early — slow kidney, eye & heart disease

Runs on data India already has

wearables + CGM + records, with no new hardware

THE ROAD AHEAD

NOW

Proof of concept

Fused model, dashboard & simulation on synthetic + open data.

NEXT

Real data

Train & validate on ShanghaiT2DM, MIMIC-IV and wearable cohorts.

THEN

Prospective pilot

A silent-mode hospital pilot measuring real lead time & utility.

LATER

Regulation & scale

Bias audits, a CDSCO pathway, clinician-in-the-loop trials.

A gentle nudge *before* the spike — not a diagnosis *after* the damage.

Thank you.

- Fuses record + wearable/CGM — both streams proven to add value
- Predicts ~50 min ahead · ROC-AUC 0.95 · 84% caught early
- Explains every alert & simulates the fix on a virtual twin
- Runs in 45 s, zero downloads — synthetic + real-data ready

Team Healthcure · IIT Kanpur